Es 504

Final Term preparation

Question:1 long Question

write four main use of the development view?

Answer:

The development view in 4+1 Architecture view model.

1) Organizing Software Component
2) Defining Module Dependencies
3) Software Configuration Management
4) Tracking proformance Maintainability.

Organizing Software Component:

Help user in managing code structure

Defining Module Dependencies:

Show interaction between different Software module.

Software Configuration Management:

Help version control and team collaboration.

Tracking performance Maintainability:

Help optimize the codebase for scalabi

Question:2

Two case (horizontal partitioning vertical partitioning) where given. Write which partitioning type applies to each case.?

Answer:

Horizontal partitioning:

Dividing tables row wise used for scability / load blancing.

Vertical partitioning:

Dividing table column wise used for performance optimization.

Question:3

In which process most number of defect will be converted with limite resources.? Also name

the software testing proc

Answer:

Process:

Th

resources

Software:

plan do and exp

Questio

server

Answe

the software document used for basic testing process.?

Answer:

Process:

that finds most defect with limited resources : Static Analysis / code Reviews

Software:

Docement for basic testing: Test plan document include (Test case, strategies) and expected result.

Question: 4 long Question

Four configuration of client server model. ?

Answer:

1 tier Architecture

2 Tier Architecture

3 Tier Architecture

N Tier Architecture

3 layer for scalability, security and performance.

Example: Enterprise app (Amazon Netflix).

Question: 5

Naming convention camelcase vs pascal case.?

Answer:

Camelcase:

1) First word start with a lowercase letter.
2) used for variable and function name.

Pascal Case:

Each word start with an uppercase letter.

used for class method name.

Question:6

Primary uses of Structure Model?

Answer:

1) Developer 2) System Architecture
3) Stakeholder

Developer:

understand structure and maintainab

System Architecture:

Manage software design.

Stakeholder:

Ensure business requirement are met.

Question:7

what is the boolean Testing technige in software testing.?

Answer:

Boolean testing technige is a white box testing method used to test logic expression by

checking
condition

Question

of re

Answer

1) Cen
4) H yb

Quest

of H

Answ

1)
2)
3)
4)
5)
6) D

checking all possible combination of condition like AND, OR and Not.

Question: 8

What are the four type of repository model in the OSI Model.?

Answer:

1) Centralized 2) distributed 3) Federated
4) Hybird Repository Model.

Question: 9

List and name the 7 layer of the OSI Model.?

Answer:

1) Application layer
2) presentation layer
3) Session layer
4) Transport layer
5) Network layer
6) Data link layer
7) physical layer

Question: 10

What is the testing phase in Software development life cycle.?

Answer:

1) Unit Testing 2) Module testing
3) Subsystem testing 4) System Testing
5) Acceptance testing 6) Aepha testing
7) Beta testing

Question: 11

Define Cyclomatic Complexity?

Answer:

Cyclomatic Complexity is a quantitative measure of the logical complexity of a program.

Question: 12

Define Inspection and testing.?

Answer:

Inspection:

Inspection can check conformance with a specification but not conformance with the customer real requirement.

Testing:

Software testing is a process of verifiying and validation whether the software product or application is working as expected or ~~not~~.

Question: 13

Define Inspection checklist.?

Answer:

Checklist of common error in a program should be developed and used to drive the inspection process.

Question: 14

What is debugging.?

Answer:

Debugging is the process of

finding and fixing error or bug in software or hardware

Question: 15

Define and explain any seven classes of software bug with example.?

Answer:

1) Memory and resource leak
2) logical Error
3) Coding Error
4) Memory over runs
5) Loop Error
6) Pointer Error
7) Boolean bugs

Question: 16

Explain Memory and resource leak and logical Error.?

Memory

memory :
but never
the funct
resource: Progr
(file / soc
closing i

Symptom

1)
2) Cr
long pe

Example

File
text
by

Memory leak and resources:

memory : Program allocate heap memory but never fresh it heap grows each time the function runs.

Resource: Program open an OS resources include (file/socket) and exist the scope without closing it.

Symptoms:

1) system sloudown

2) Crashes that occour "randomly over a long period of time.

Example:

```
void foo() {
int *p = new int [100]; // alloc
// ... use P..
} // lost reference here leak
void read () throws ! O Exception {
File input stream = new Fileinput stream(date.
text); // acquired
byte [] buf = in read All bytes
```

Logical Error:

Logical Error occurs when the code is syntactically correct but does not do what you expect it to do.

Symptoms:

1) The code misbehaving in a way that is not easily explained.
2) Result are the opposite of what is expected.

Example:

```
Int sagure (x)
{
return x*2 // Should be x*x
}
```

Question:17

Explain Coding error and Memory over runs with example.?

Answer:

Coding Error:

A coding error is a simple problem in writting the code.

Symptoms:

1) Unexpected error in black box testing
2) Code warning

Example:

```
Cout << "Hello" // Missing smi colon
```

Memory over runs:

A memory overruns occurs when you see memory that does not belong to you. This can be overstepping an arry boundary by copying a string that too being.

Symptoms:

1) Program crashes quite regularly after a given routine is called memory.
2) Trashed variable or memory block.

Example:

```
char arr[5] = "Hey";
arr[5] = '!'; // Memory over runs
writing outside arry bound
cout << arr << endl;
```

Question: 18

Explain loop error and pointer Error with Example.?

Answer:

Loop Error:

Infinite loop, off by one loop and improperly existed loop.

Symptoms:

1) If you program simply locks up.

2) Off by one loop error are quite often seen in processes that perfom calculati

Example:

```
int i =0;
while (i != 10) {
cout << i << end
i+=2 ; // i will never be
```

exactly 10

```
}
```

Question No.33

Differentiate between Black box testing and white box testing.

Answer:- (Page 198)

- In this type of testing, a component or system is treated as a black box and it is tested for the required behavior. This type of testing is not concerned with how the inputs are transformed into outputs.
- As opposed to black box testing, in structural or white box testing we look inside the system and evaluate what it consists of and how is it implemented.

Question No: 34 (Marks: 3)

Question No: 31 (Marks: 2)

Differentiate between Architectural Design and System Architecture in one line.

Answer:-

Architecture faces towards strategy, structure and purpose, towards the abstract while Design faces towards implementation and practice, towards the concrete.

Question No: 59

WHITE BOX TESTING?

Answer:- (Page 202)

In white box testing we test the structure of the program. In this technique the test cases are written in
to cover different possibilities in code.

Write unit testing principles?

In unit testing, developers test their own code units (modules, classes, etc.) during implementation.

Normal and boundary inputs against expected results are tested.

Thus unit testing is a great way to test an API.

What are the six elements that are present in every computer-based system?

1. Hardware
2. People/user
3. Data
4. Procedure
5. Connectivity

What is called self documenting code?

A self documenting code is a code that explains itself without the need of comments and extraneous

documentation, like

Define Modularity?

Modularity is a tool that can help us in reducing the size of individual functions, making them more readable.

What is meant by Software Debugging?

Debugging techniques are the only mechanism to reach at the code that is malfunctioning.

Past Paper

CS 504

Final term Prepation

Question:1

Assume the code given below.

```
for (i=0 ,i <N ++) {// 1
if (condition 1)
// do something here //2
 else
// do something //3
 //Something here //4
 }
```

Determine the number of possible path that can traversed the above code given the value of variable N is 2. Also identife those path.

Solution:

Number of path: There are 4 possible path.

Summary:

Number of possible path 4

Paths:

1) 1 → 2 → 4 → 1 → 2 → 4
2) 1 → 2 → 4 → 1 → 3 → 4
3) 1 → 3 → 4 → 1 → 2 → 4

Date: _____

4) 1 → 3 → 4 → 1 → 3 → 4

Question: 2

Consider the following code fragment while a

{

while b

c

d

}

If you were to test this code. what would be the test technique to adopt.?

Solution:

To test given code fregment, you should consider using path testing (also known as basis path testing), which is a white box testing technique. Path testing ensure that all possible path through the code are tested at last once.

```
while a {
while b {
(;
```

Question: 3

Consider the following value and calculate cyclomatic complexity.

Number of edge = 8

Number of nodes = 6

Solution:

Number of edge = 8

Number of nodes = 6

Cyclomatic Complexity, V(G), for a flow graph G is defined as.

V(G) = E - N + 2

C(G) = 8 - 6 + 2 = 4

Path to be tested

Path 1: 1 - 6

Path 2: 1-2-3-4-5-1-6

Path 3: 1-2-4-5-1-6

Path 4: 1-2-4-2-3-4-5-6-1

Question: 4

Read the following statement carefully and mark it as correct or incorrect and in case of incorrect statement

Date:

make it correct.

"In the telecommunication industry loss of service is know as a lack of service."

Solution:

It is false

Correct statement is

In the telecommunication industry, loss of service is know as outage.

Question: 5

In Krutchen architectural view model, who are the primary user of concurrency view.?

Solution:

Concurrency view

1) Performance engineers,

2) integrator 3) Tester

Date:

File 2

CS 504

Final Term Preparation

less

service.?

Question:6

Identify the class object and relationship from the follow scenairo "Ali has a fruit basket.?

Solution:

When a class is formed as a collection of other classes, it is called aggregation relationship.

Ali has a fruit basket is a aggregation.

Question:7

white box coverage explain three types.?

Answer:

1) Whit box coverage focuses on internal code structure such as branches, condition, loop, and paths.

2) It help in detecting logical errors, unreachable code, and Security vulnerability at the code level.

3) Common technique under the white box Testing include statement coverage decision

coverage and path coverage.

Question: 8

Exception handling write three level.?

Answer:

1) Detection 2) Propagation

3) Handling.

Question: 9

Software Architecture Observer pattren write name.?

Answer:

1) Subject 2) Observer

Question: 10

Asynchronous, Synchronous and destory Message symbol.?

Answer:

Synchronous: ——▶

Solid line with filled arrowhead

Asynchronous: ——▷

Solin line with open arrowhead

Destory Message: X

lifeline.

Question: 11

Do yo
effect of a fun
difficult during
activities.?

Answer:

Yes, I
can make mainte
difficult S
1) Modifying
2) Writing or
3) Printing to

Question: 12

As
are modeling
to the func
are mandato
as E.O.?

Answer:

1) Da
2) Data
Bound

Date:

Question:11

Do you agree that a side effect of a function can make things difficult during maintenance or debugging activities.?

Answer:

Yes, I agree side effect in a function can make maintenance and debugging more difficult **Side effect**

1) Modifying a global variable
2) Writing or reading from a file
3) Printing to the console 4) Updating a database

Question:12

As a requirement engineer, you are modeling business requirement according to the function point analysis. Which condition are mandatory for a function to be categorized as E.O.?

Answer:

1) Data organized from inside the system
2) Data is send outside the application Boundary.

3) Processing Logic is included

Question: 13

In which two ways the shift operator (>>) can be used in C and C++.?

Answer:

1) As a bitwise right shift operator to shift bits to the right. (in both C/C++)

2) As a Stream extraction operator to take input from the user using cin (C++ only)

Question: 14

What is the general style variable guideline for naming generic variable in a program.? Give an example.

Answer:

Generic variable should have meaningful clear names, following a consistent style like "camel class" or "snake class". For temporary variable (like loop counters) Single letter like I, j are okey.

Example:

Good: userCount, MaxValue

loop variable: i, j

Question: 15 What are the main Three tier architectural.?

Answer:

1) Presentation layer 2) Application layer 3) Data layer

Question: 16

What are the four component of GoF design pattern.?

Answer:

1) Creational Pattern 2) Structural pattern 3) Behavioral pattern 4) concurrency pattern

Creational pattern:

used for creating object in a flexible way.

Structural Pattern:

Date:

Deal with object and classes composition.

Behavioral Pattern:

Handle communication between object.

Concurrency Pattern:

Used for multi threading and parallel execution.

Question:17

What is a Software bug.?

Answer:

A software bug is an error, flaw or fault in a computer program that causes it to produce incorrect and unexpected result or to behave in unintended ways. Bug can occure due to mistake in the code design or logic.

Question:18

Following is the code written to calculate the areea of a circle but it is difficult to understand because of name identifier.?

Date:

Answer:

The

The name unclear. Usi

and area

and unders

Question:1

names.?

Answer:

meaningf

area wh

that the

a circle

Questi

architect

Answe

model.

Date:

Answer:

The code is hard to understand because the name of variable and function are unclear. Using meaningful name like radius and area makes the code easier to read and understand.

Question: 19

Rewritten code with better names.?

Answer:

Rewritten code uses clear and meaningful names like radius, pi, and area which make it easy to understand that the program calculate the area of a circle.

Question: 20

Write the name of krutchen architectural view model.?

Answer:

Krutchen 4+1 Architectural view model.

Date:

Question: 21

"Int factorial"; write it as ANSI Standrad.?

Answer: American national standrad institute ANSI standrad requires variable declarations inside a function.

```
int factorial = 0;
```

Question: 22

In telecommunication system lack of service is due to lack of service." In this statement correct.? if not correct it.?

Answer:

Correct statement:

In a telecommunication system lack of service is due to network congestion, resource allocation issues or system failure.

Question: 23

Enlist the type of interaction diagram.?

Date:

Answer:

Two ty

1) Sequence di

Sequence diag

a time sequ

Collaboration

is a Emphasi

Question: 24

A

with 9 Edg

Complexity.?

Answer:

Cycl

$v(G) = E -$

Number

Number

V

Cyclom

Answer:

Two types of interaction diagram.

1) Sequence diagram 2) Collaboration diagram

Sequence diagram:

Show object interaction in a time sequence.

Collaboration diagram:

Collaboration diagram is a Emphasize object relationship.

Question:24

A flow graph was given with 9 Edge and 6 Nodes Find cyclomatic Complexity.?

Answer:

Cyclomatic Complexity Formula

$$v(G) = E - N + 2$$

Number of Edge = 9

Number of Nodes = 6

$$v(G) = E - N + 2$$

$$9 - 6 + 2 = 3 + 2$$

Cyclomatic Complexity = 5 Ans

Date:

Question: 25

write three alternative of "is" prefix.?

Answer:

Alternative for "is" prefix in variable, method name.

1) has
2) can
3) Should.

Question: 26

what is error handling.?

Answer:

Error handling is a technique used to manage error in a program so that it doesn't crash. It uses block try and catch to detect and handle errors and provide proper message to the user.

Question: 27

which diagram are used to present dynamic behavior of the system?

Answer:

1) Sequence diagram
2) State machine diagram
3) Activity diagram

Sequence diagram:

Show object interaction over time.

State machine diagram:

Represent change in state of an object.

Activity diagram:

Depicts workflow or the control flow operation.

Question: 28

Explain how can the ~~use~~ size of the function in a program affect program maintainability.?

Answer:

Larger function are header to read, understand test and debug. They increase complexity making maintance difficult. Smaller reuse named function improve readability, reuse, and ease of update, enhancing maintainability.

Date:

Suppose that you write a C program in which you declare a pointer to dynamically allocate memory to store the ages of 10 student. But you forget to delete the pointer. In doing so, what type of bug will occur in the program?

Answer:

If you forget to delete the pointer a memory leak will occur. The allocated memory won't be freed, causing inefficient memory usage.

Question: 30

Suppose that software testing phase has been completed and all the defects have been discovered. What is the next step for after completion of test phase.?

Answer:

After the testing phase is completed and defects are discovered, the next step is defect fixing (debugging) followed by development or release of the software.

Date:

Question: 31

Bit

lack of portab

Answer:

Bit fiel

Compiler handle

arding differe

platform. This

across system

Question: 32

Co

1) A hard driv

files. whene

or file and

is deleted a

from where

2) using fcter

at least

Answer:

u

container

Date:

Question: 31

Bit field do suffer from lack of portability between platform.? why

Answer:

Bit field lack of portability because compiler handle size, alignment, and bit ording differently on different (problem) platform. This can lead to inconsistent behavior across system.

Question: 32

Consider the following scenario.

1) A hard drive contain folder which contain files. whenever use right click on the folder or file and presses delete button the file is deleted and is sent to recycle bin, from where it can be deleted permanently.

2) using peter coad methodology, identify at least two container objects.?

Answer:

Using peter coad methodology, two container object in this scenario.

Date:

1) Hard disk drive 2) Folder

Hard disk drive:

Contain folder

Folder:

Contain files.

Question: 33

Name at least three fault clases which must be specified while preparing an inspection checklist.?

Answer:

1) Initialization faults
2) Data reference faults
3) Control faults

Question: 34

How does profiling work.?

Answer:

Profiling works by analyzing a program ruline behavior, such as function calls memory usage and execution time. It helps identify performance bottlenecks by data during program execution.

Date:

Question: 35

R

General ran

1) int file

2) Const int

3) int get

4) Templat

Answer:

1) // us

2) // Prefix

3) // use

4) // Add sp

Question:

a comb

Those

Answer

1) Arc

3) Data

5) Inte

Date:

Question: 35

Rewrite the following code General naming convention for C++.

1) int fileprefix
2) Const int max iteration
3) int getmaxiteration () {...}
4) Template < class t >.

Answer:

1) // use camel class for variable name
2) // Prefix constant with k and use Pascal class.
3) // use camel class for function name.
4) // Add space and capitalize templete parameter.

Question: 36

Software design phase is a combination of number of activities. List those activities?

Answer:

1) Architectural design 2) High Level design
3) Detaild Design 4) Data design
5) Interface Design 6) Component Design

Date:

Question:37

Compute the cyclomatic complexity of the code below.

```
If (a>10)
    C++
    else
```

Answer:

CC = E - N + 2

CC = Number of decision points + 1

If (a > 10) -> 1

1 + 1 = 2 Ans,

Question:38

Do you agree that a side effect of a function can make things difficult during maintance or debugging activities.?

Answer:

Yes Side effect can make maintaince and debugging header because they create unexpected changes outside the function, making the program behavior less predictable.

1- Software Engineering is result of software crisis.

2- Dcentralization.

3- Architechtural styles: 3 types

- Data Center architecture
- Client Server Architecture
- Layerd architecture

4- AND or OR are Relational operation.

5- 2/3 cost incurs during maintainance Phase.

6- Function oriented methodology is also known as Action Or

7- Umberalla activities:

- ✓ Risk manggement
- ✓ Software Quality Assuran
- ✓ Software Contiguous Management

8- Equivalence classes help you designing test cases.

9- Non-Functional requirement: Inspection can't check ← ch

- White Box → test the structure of the progra
- Black Box →

Sequence Diagram: steps of sequence

- Synchronous message mean callevent. Full arrow

The caller wait for the called routine to cplte its operation before moving forward -

- Asynchronous message mean signal, denoted by half arrow

creat new object, new thread, com with object on

thread, that is already running

- Create & destroy message another

An object may create object via a <<create>> message

An object may destroy itself via a <<destroy>>

Collaboration diggram: 2 main component:

- object
- messages. → respect to organizational units

- Message are numbered and can have loops.

Sequence → flow

are best for

- Flow of time
- ~~Difficult~~ Easily Understand

Why is architecture important?

- Mutual communication
- Early design decision
- Reusable

Recusable abstraction of a system

Collaboration diagram

- Static object connection
- Difficult undrestaing of sequence steps
- very quickly
- Very complex

Naming Convection

1- General naming convection for C++ , Java

Rules-

1- Name representing types must be nouns & written in mixed case starting with upper case. (Name must be nouns written in mixed & start in —
Line, FilePrefix

2- Variable names in mixed case with lower case.

Line, filePrefix

3- Naming representing Constant: must be all uppercase using underscore to separate words.

MAX_ITERATION, COLOR_RED

4- Name representing methods & functions should be verbs &. written in mixed case with lower case.

getName(), computeTotalWidth()

5- Names representing template types in C++ should be single Upper case letter.

template <class C ,class D >.....

6- Global variables in C++ should be referred to by using the :: operator.

7- Private class member should have — suffix.

Private int length;

8 Abbrivation & Acronyms should not be uppercase when used as names:

Html

9 Generic Variable should have same name as the its ty

(Pic topic)

Dua for Success

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا وَرِزْقًا طَيِّبًا
وَعَمَلاً مُتَقَبَّلاً

Allahumma inni As'aluka ilman Nafian wa Rizqan Thayyiban
wa Aamalan Muttaqabbalan

O Allah, I ask You for beneficial knowledge, Goodly provision,
and Acceptable deeds

[Sunan Ibn Majah]

Exam ki Kaamyaabi ki Dua

فَاِنَّ حَسْبَكَ اللّٰهُ ؕ

هُوَ الَّذِیْۤ اَیَّدَكَ بِنَصْرِهٖ وَبِالْمُؤْمِنِیْنَ

سورہ انفال

Dua : Fa inna hasbakallahu huwal lazi ayyadaka binasrihi wa bil mu'mineen

(Al Quran 8.62)

Tareeka : Imtihaan se pehle Awwal Aakhir me durood sarif padh kar 7 Martaba ye Dua padhe..

استغفار کے فوائد!

(1) گناہ معاف ہو گئے۔ (سورۃ نوح : 10)

(2) بارش ہوگی۔ (سورۃ نوح: 11)

(1) اولاد نصیب ہوگی (سورہ نوح: 12)

(4) مال ملے گا (سورۃ نوح : 12)

(6) نہریں جاری ہونگی (سورۃ نوح : 12)

(7) خوش حالی والی زندگی نصیب ہوگی (سورہ ہود: 3)

(8) طاقت اور قوت ملے گی (سورہ ہود: 22)

(9) مبارکباد ہے اس شخص کے لیے جس کے نامہ اعمال
میں استغفار زیادہ ہو۔ (ابن ماجہ)

(10) جو پابندی سے استغفار کرے گا تو اللہ اسے ہر غم اور ہر
تکلیف سے نجات دے گا ہر تنگی سے نکلنے کا راستہ پیدا فرمائے گا
اور اسے ایسی جگہ سے رزق دے گا، جہاں سے اسے وہم و گمان بھی نہ ہوگا